PHYS30762 Object Oriented Programming in C++

Analogue Circuits Report

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Abstract

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1. Introduction

This program allows users to build analogue circuits using resistors, capacitors, and inductors (which are all classes derived from an abstract base `Component` class). A `Circuit` class can be set to be series or parallel, and any number of components can be added to it, which is then stores as a vector of components (specifically a vector of unique pointers to components). Since the `Circuit` class is inherited from the `Component` class itself, it is possible to add series circuits to parallel circuits, or vice versa, and therefore construct circuits of any topology.

In the main file of this program, a *library* of components is created, which stores different types of resistors, capacitors and inductors. These components are validated through the use of validation files, such that if the user tries to initialise a component with properties which do not match those specified in the validation file, an error message will be returned.

2. Code design and implementation

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3. Results

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4. Discussion and conclusions

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